

Module 4: Intermediate Statistics and Inference — Summary

What this module is about. The normal distribution gives us a way to describe where individual values fall relative to a mean. A z-score measures distance from the mean in standard-deviation units. Statistical inference asks a different question: what can a sample tell us about a population? A confidence interval combines a sample estimate with a margin of error to give a plausible range for a population value.

For an approximately normal distribution:

- about 68% of values lie within 1σ of the mean;
- about 95% lie within 2σ ;
- about 99.7% lie within 3σ .

The remaining percentage is split equally between the two tails when the distribution is symmetric.

Example 1: Apple weights

Apple weights are approximately normal with mean $\mu = 180$ g and standard deviation $\sigma = 15$ g.

- One standard deviation from the mean is 180 ± 15 , or 165 to 195 g, so about 68% of apples lie there.
- Two standard deviations from the mean is 180 ± 30 , or 150 to 210 g, so about 95% lie there.
- About 5% lie outside that interval, so about 2.5% weigh more than 210 g.

For an individual value x ,

$$z = \frac{x - \mu}{\sigma}.$$

A positive z-score means the value is above the mean; a negative z-score means it is below the mean. To recover a value from a z-score,

$$x = \mu + z\sigma.$$

Because z-scores use standard-deviation units, they allow comparisons across different scales.

Example 2: Comparing two standardized tests

Student A scores 520 on a test with mean 500 and standard deviation 100:

$$z_A = \frac{520 - 500}{100} = 0.20.$$

Student B scores 28 on a test with mean 21 and standard deviation 5:

$$z_B = \frac{28 - 21}{5} = 1.40.$$

Student B performed farther above the mean relative to the spread of that test.

A **parameter** describes the population. A **statistic** is computed from a sample and is used to estimate the parameter.

If an estimate is reported as $E \pm M$, then the corresponding interval is

$$[E - M, E + M].$$

A larger sample generally produces a smaller margin of error. In the simplified model used here,

$$M \propto \frac{1}{\sqrt{n}}.$$

Therefore, multiplying the sample size by 4 approximately cuts the margin of error in half.

Example 3: A label claim and a sample interval

A cereal label claims a population mean of 230 calories. A sample gives 243 ± 8 calories, so the interval is

$$[243 - 8, 243 + 8] = [235, 251].$$

Because 230 is outside this interval, the label claim is outside the plausible range given by this sample and margin of error.

Module 4: Intermediate Statistics and Inference — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Useful formulas

$$z = \frac{x - \mu}{\sigma}, \quad x = \mu + z\sigma, \quad \text{confidence interval} = \text{estimate} \pm \text{margin of error}.$$

Empirical Rule: 68%, 95%, 99.7% within 1, 2, 3 standard deviations.

1. Orange weights are approximately normal with mean 200 g and standard deviation 10 g.
 - (a) About what percentage of oranges weigh between 190 g and 210 g?
 - (b) About what percentage weigh more than 220 g?
 - (c) Between what two weights do the middle 95% of oranges fall?

2. Student A scores 78 on a test with mean 70 and standard deviation 4. Student B scores 540 on a different test with mean 500 and standard deviation 50.
 - (a) Find each student's z-score.
 - (b) Which student performed better relative to the distribution of scores on that student's test?

3. A consumer group tests bottles of a drink. The sample mean is 102 mg of a nutrient per serving, with a reported margin of error of ± 5 mg. The label claims a population mean of 95 mg.
 - (a) Construct the confidence interval.
 - (b) Is the claimed mean of 95 mg inside the interval?
 - (c) In this problem, is 102 mg a population parameter or a sample statistic?

4. A study uses a sample of $n = 150$, and its margin of error is approximately proportional to $1/\sqrt{n}$. About

what sample size would be needed to cut the margin of error in half? Show how the square-root relationship determines your answer.

5. A distribution has mean $\mu = 50$ and standard deviation $\sigma = 8$. An individual value is $x = 66$.
- (a) Find the z-score of 66.
 - (b) Is the value within two standard deviations of the mean?
 - (c) Does this z-score answer a question about an individual observation or about a population parameter?

Module 4: Intermediate Statistics and Inference — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. Commute times are approximately normal with mean 35 minutes and standard deviation 8 minutes. Use the Empirical Rule to estimate the percentage of commutes between 19 and 51 minutes.
2. On one standardized test, a student has $x = 84$, $\mu = 72$, and $\sigma = 6$. Find the z-score and state whether the score is above or below the mean.
3. A survey reports 47% support for a proposal with margin of error $\pm 6\%$. Construct the interval. Is 55% inside it?
4. A sample mean of 31.2 cm is computed from 40 randomly caught fish. Is 31.2 cm a parameter or a statistic? What population quantity is it intended to estimate?
5. If a sample size is multiplied by 9, by what factor does the simplified $1/\sqrt{n}$ model predict the margin of error will change?